ENGR 312: Automatic Control Systems
Homework Assignment #6
(30 pts max.)
Professor Blaise

Read Chapter 8, sections 8.1-8.5, and 8.8-8.9 of the textbook. And do the following problems.

1) Given the linear time-invariant system described by the differential equations below, use the **state-space approach** to write the state equations and output equations **in vector-matrix form**.

a) $$2\frac{d^3y(t)}{dt^3} + 3\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 2y(t) = r(t)$$ [6 pts]

b) $$\frac{d^4y(t)}{dt^4} + 1.5\frac{d^3y(t)}{dt^3} + 2.5\frac{dy(t)}{dt} + y(t) = 2r(t)$$ [6 pts]

2) Given the systems defined by the transfer functions shown below, use **MATLAB** to convert from transfer function to **state-space representation in vector-matrix form**. (**Attach a screenshot of the Matlab commands/screen outputs you used to solve this problem**).

**** PLEASE DO NOT LET AI WRITE THE MATLAB COMMANDS FOR YOU. CONSULT YOUR CLASSNOTES AND WRITE THE COMMANDS YOURSELF! ****

a) $$\frac{Y(s)}{U(s)} = \frac{s^2 + 3s + 7}{(s+1)(s^2 + 5s + 4)}$$ [4 pts]

b) $$\frac{Y(s)}{U(s)} = \frac{-25}{s^2 + 4s + 25}$$ [4 pts]

3) Given the state-space representations of the systems shown below, use **Matlab** to convert from state-space to **transfer function representation**. (**Attach a screenshot of the Matlab commands/screen outputs you used to solve this problem**).

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a) [6 pts]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

b) [4 pts]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -4 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u$$

$$y = [1 \quad 1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$